2025 44th International Conference of the Chilean Computer Science Society (SCCC) | 979-8-3315-9716-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/SCCC67219.2025.11420704

Improving Emergency Triage Accuracy with Gradient Boosting Algorithms and NLP-Derived Features

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***Abstract*—In Chile, patient care in emergency departments is organized through a process known as categorization or triage, which assigns priorities according to the severity of each case. This procedure is performed by nurses using predefined criteria; however, such evaluations can be affected by subjectivity and human error, which can have repercussions for patients with severe conditions who may receive an incorrect classification, potentially delaying the necessary emergency care and generating operational bottlenecks that negatively impact the quality of care.**

This study addresses this problem using Natural Language Processing (NLP) and Machine Learning techniques. Using the Word2Vec model, the free text obtained in triage is transformed into numerical vectors, thus facilitating its incorporation into classification algorithms. The comparative performance of different algorithms, including CatBoost, XGBoost and LightGBM, on the same dataset is evaluated. The metrics analyzed include accuracy, weighted F1-score, and cross-validation using the K-fold method.

The results reveal that, by using NLP in conjunction with machine learning models based on gradient boosting algorithms, a correct prediction of the different severity categories is achieved, with the CatBoost algorithm yielding an accuracy of 0.733 and a weighted F1-score of 0.731. Although the different models showed similar performances, it was clinically relevant to prioritize those with lower false-negative rates to accurately identify the severity of each patient.

***Index Terms*—Artificial intelligence; emergency department; word2vec; Gradient Boosting Algorithms.**

I. Introduction

Emergency departments (EDs) are a vital pillar of healthcare systems worldwide. They are the main gateway to the healthcare network and represent the initial point of contact in critical or highly complex clinical situations. According to [16], the saturation of emergency departments is due to multiple causes and has significant effects on patient care. Similarly, [23] highlights that demographic changes, such as an aging population and the growing prevalence of chronic diseases, have contributed to a sustained increase in the demand for emergency services.

In the Chilean context, [9] emphasizes these limitations of primary care in efficiently resolving health problems, which leads patients to seek emergency services. In addition, [5] argues that many patients perceive emergency services as faster and more effective than other healthcare alternatives, reinforcing their role as the preferred point of entry into the system.

To efficiently manage increasing patient demand and complexity, triage systems have been adopted in multiple countries. Among the most widely used are the Australian Triage Scale [2], the Canadian ED Triage and Acuity Scale [3], the Manchester Triage System (MTS) [20], and the Emergency Severity Index (ESI) [12], the latter being the model currently implemented in Chile's public hospitals. The ESI classifies patients into five levels (C1-C5) based on clinical severity and the resources required for their care, aiming to optimize the prioritization of cases and the allocation of medical resources.

Despite its widespread adoption, triage accuracy remains a challenge. Several studies have reported overtriage rates ranging from 4% to 15%, and undertriage rates ranging from 4% to 22% [11], which can compromise both patient safety and system efficiency. Factors such as human variability, care pressure, and subjectivity in case interpretation directly influence the quality of clinical decisions [10].

In this context, artificial intelligence-based support tools present themselves as a promising alternative to improve the consistency and accuracy of the triage process. In particular, the use of natural language processing (NLP) techniques enables the exploitation of information contained in clinical notes written by nursing staff at the time of patient admission.

This paper proposes an innovative approach that predicts the triage severity category from the descriptive texts of the reasons for consultation recorded in the emergency department. For this purpose, the word2vec technique [21] is used to represent the texts semantically, and three supervised learning algorithms based on decision trees and gradient boosting — XGBoost [4], CatBoost [24], and LightGBM [18] — are compared in a highly unbalanced multiclass classification scenario. Additionally, oversampling and Bayesian hyperparameter optimization techniques are employed to enhance the robustness and generalization of the models.

In contrast to previous research that has focused primarily on binary predictions or the use of structured variables extracted from electronic clinical records [8], [27], this study demonstrates the feasibility of using only the free text of the reason for consultation to accurately predict the triage category according to the emergency severity index (ESI). Using nat-

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ural language processing techniques, such as word2vec [21], and advanced gradient boosting algorithms, we were able to capture implicit clinical information in the medical notes at admission. Recent studies have shown that the use of free text can improve patient stratification in emergency settings [6], [7]; however, no research has been identified that applies this methodology using real data from a highly complex emergency department in Chile. In this sense, the results of this work demonstrate the potential of the proposed approach to support clinical decision-making in local scenarios, contributing to a more efficient, accurate, and equitable emergency system.

II. STATE OF THE ART

The authors of the scientific article "Predicting drug-target binding affinity based on power graphs and Word2Vec" [17] apply Word2Vec to represent protein characteristics in order to understand local chemical information and amino acid sequences. Word2Vec is used within a model to encode amino acid sequences into an embedding matrix. Word2Vec also performs the task of creating biological words composed of three amino acids. Each word is then mapped to an embedding vector, and these words are fed into a convolutional neural network to extract information and learn from their features. To compare the effectiveness of Word2vec, it was compared with traditional One-hot encoding. One-hot proved to be worse, performing 3.6% worse on the consistency (CI) metric and having a 38% increase in the mean quantum error.

The study by Halwani et al. (2025) [14] demonstrated the significant potential of machine learning models to improve the accuracy of pediatric triage based on the Canadian Triage and Acute Care Scale (CTAS) framework, achieving an overall accuracy of over 95.5%. Gaussian Naïve Bayes (GNB) achieved the highest overall accuracy of 98.4%, highlighting the ability of simple models to achieve overall predictive performance. However, the research concluded that Support Vector Machine (SVM) and Gradient Boosting (Random Forest and LightGBM) models offered a better balance, achieving the highest performance in terms of consistency of accuracy and recall across all CTAS levels. Because GNB showed limitations in the extreme categories (CTAS 1 and CTAS 5) due to its assumption of feature independence, Random Forest and LightGBM models were robust in predicting CTAS 3 and CTAS 4. The study also highlights that the lack of interpretability in complex models such as SVM and Gradient Boosting remains a substantial barrier to clinical adoption.

Finally, Uzeyir Gurbanli, author of "Application of the Word2Vec algorithm for clinical diagnosis determination" [13] explores the use of Word2Vec for processing and refining clinical diagnoses, especially diagnoses of cardiovascular diseases, because it is practical to use these NLP (Natural Language Processing) technologies in the medical field due to the large amount of semantically rich text that allows for more precise diagnosis determination and identification. The author also concludes that Word2vec and its proposed methodology can be used in any language, due to its vector representation approach. Based on the tests carried out, the algorithm yields a cosine similarity of 0.591 in detecting asthma diagnoses and 0.578 in detecting hypertension diagnoses.

III. MATERIALS

A. Data set

The data set used in this study corresponds to the records of patients attended in the Hospital Regional de Talca emergency department between 2016 and 2017. On average, this highly complex hospital attends 120,000 patients annually in its emergency department. Of this total, approximately 47% corresponds to adult care, 33% to child patients, 18% to obstetric-gynecological consultations, and 3% to dental care.

To ensure consistency in clinical and operational analysis, we filtered the dataset to include only records corresponding to adult patients and dental consultations. We excluded all records for patients under 15 years of age and those related to obstetrics and gynecology services, as these groups exhibit different patterns of behavior than the adult population in emergency departments.

The dataset classifies patients into five triage categories: C1 (vital emergency), C2 (severe emergency), C3 (moderate emergency), C4 (minor emergency), and C5 (non-emergency). Fig. 1 illustrates the relative frequency of each category. Categories C2 and C3 comprise approximately 90% of all visits, reflecting a significant burden of care associated with non-critical but time-sensitive cases. In contrast, categories C1 and C5 represent less than 2% of the cases, consistent with their clinical nature as infrequent but extreme events (vital emergencies or non-emergencies).

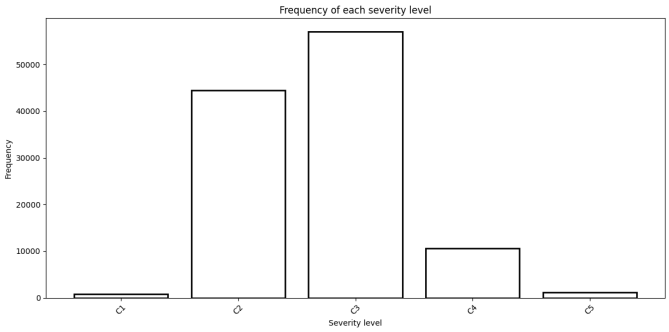


Fig. 1. frequency distribution of patients by severity category according to the Emergency Severity Index (ESI) system at Hospital Regional de Talca, between 2016 and 2017.

The percentage distribution of triage categories in the period analyzed was as follows: C1 (1%), C2 (39%), C3 (50%), C4 (9%), and C5 (1%). This marked imbalance in class frequency is critical for the predictive models developed in this study and will be addressed in detail in later sections.

Each patient record includes the reason for the consultation in free text, summarizing the patient's condition and the reason for their care. This field has a high semantic density and specialized clinical vocabulary, where acronyms, abbreviations, and spelling variants are common. This characteristic of the reason for consultation, which is not standardized, poses

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challenges for normalization and linguistic disambiguation, which directly impacts automatic classification.

IV. Methods

A. Data Preprocessing

To achieve an effective vector representation using Word2Vec, a preprocessing phase is required for the dataset. The main goal is to clean, normalize, and structure the *Motivo_cons* column, which contains the description of the reason why the patient attended triage. This process aims to reduce noise and improve the semantic quality of the corpus. The applied procedure is detailed below:

- **Text normalization**: We converted all characters to lowercase and removed punctuation marks, numbers, diacritics, and non-alphabetic characters. We also replaced parentheses and eliminated unnecessary or duplicate spaces.
- **Acronym replacement**: To address the frequent use of medical abbreviations, we created a custom dictionary with approximately 150 acronyms and used regular expressions to replace them with their expanded forms automatically.
- **Lemmatization**: We applied Lemmatization using the `es_core_news_sm` linguistic model from the `spaCy` library, reducing each word to its canonical or grammatical root form.
- **Stopword removal**: We removed functional terms with no significant semantic value.
- **Tokenization**: Finally, we segmented the text into relevant words (tokens) and discarded tokens with fewer than three characters or deemed irrelevant.

This preprocessing resulted in a structured and semantically enriched dataset suitable for input into the vectorization process.

B. Data Vectorization with Word2Vec

After cleaning and tokenizing the corpus, we performed the word vectorization using the Word2Vec model [21], which transforms each word into a continuous vector in a multidimensional semantic space. We trained the model using the Skip-gram architecture, as it has been shown to perform better on large corpora [22].

To optimize the hyperparameters of Word2Vec, we employed the metaheuristic algorithm Aquila Optimizer [1]. This swarm intelligence-based technique allowed us to explore the search space of the following hyperparameters efficiently:

- vector_size: We explored values between 50 and 1000 for the dimensionality of the semantic vectors (literature commonly recommends values from 100 to 1000 [25]).
- window: We evaluated context window sizes ranging from 2 to 10 [19].

We defined the objective function to maximize the average similarity among the three words closest to the most frequent word in the corpus, using the `cosine similarity` metric inherent to Word2Vec [21]. After training the model with the optimal parameters, we evaluated its semantic quality.

Model Evaluation: To validate the semantic coherence of the model, five representative clinical groups were defined (cardiology, neurology, infectious diseases, traumatology, and psychiatry), each with an associated set of words. These words were projected into a two-dimensional space using the **t-SNE** technique, allowing the visualization of how they cluster semantically based on the model's learned knowledge.

Fig. 2 shows a two-dimensional representation generated with t-SNE from the semantic vectors obtained using the Word2Vec model trained on texts of reasons for consultation in an emergency department. Each point represents a key word extracted from the corpus, and its color corresponds to a manually assigned clinical category: cardiovascular (red), neurological (blue), infectious (green), traumatic (purple), and psychiatric (orange). The observed spatial clustering between terms suggests that the model successfully captures consistent semantic relationships. In addition, a semantic entropy (S) value is observed for each cluster, which quantifies its spatial dispersion. Low values of S indicate high semantic cohesion between words in each category, which supports the quality of the learned embedding.

C. Predictive Models

After training and optimizing the hyperparameters of the Word2Vec model, we need to perform class balancing, as there is a significant imbalance in the dataset.

As a strategy to address the class imbalance problem, we adopted a hybrid approach, combining SMOTE oversampling with the use of custom class weights during training.

- SMOTE Oversampling: They employed an oversampling strategy to increase the number of samples from the minority classes artificially. This approach allowed us to preserve valuable information that might otherwise be lost due to class imbalance. However, we also took care to avoid excessive duplication, which can introduce noise into the learning process. To address this, we employed the Synthetic Minority Over-sampling Technique (SMOTE) [26], which generates new synthetic instances instead of replicating existing ones. We configured SMOTE with a custom sampling strategy tailored to our dataset. Specifically, we set an upper limit of 10,000 synthetic samples for the least represented categories and 15,000 for the intermediate category, ensuring that the oversampling process maintained a proportionality among classes.
- Class weighting and proportional balance: The other component of this hybrid approach involved leveraging the capabilities of the learning algorithms (XGBoost, LightGBM, and CatBoost) to manage class imbalance through custom class weighting [15]. Each of these gradient boosting algorithms provides a mechanism to assign different weights to each class during training, which allows the model to compensate for the imbalance present in the dataset. We assigned weights inversely proportional to the frequency of each class. This strategy ensured that minority classes—such as C1 and C5—had

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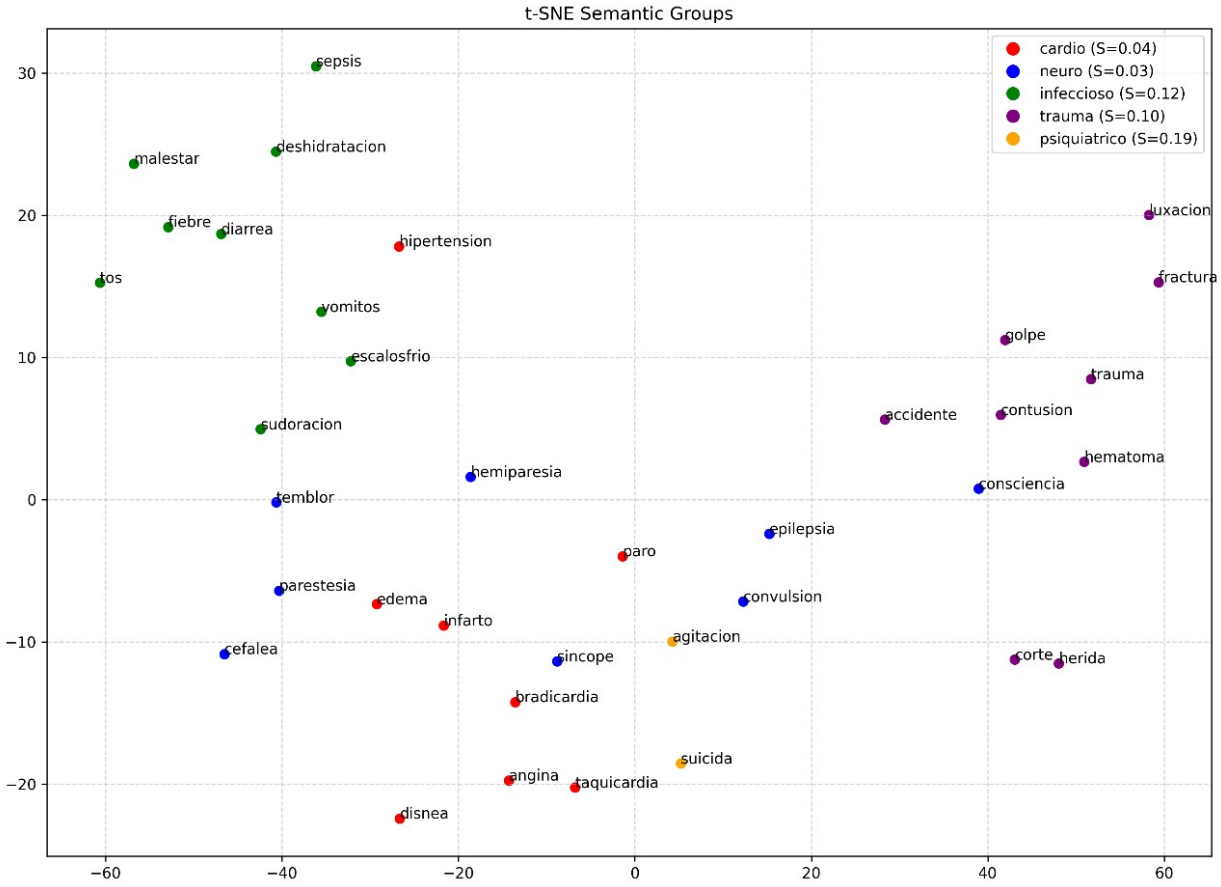


Fig. 2. Semantic clustering using t-SNE applied to Word2Vec vectors optimized with Aquila Optimizer.

a greater impact on the loss function, thereby guiding the model to better discriminate against these underrepresented categories. By doing so, we aimed to reduce the bias toward majority classes and improve the model's ability to generalize across all triage categories.

They applied a hold-out validation strategy after completing the balancing and class weighting procedures. Specifically, we split the dataset into 80% for training and 20% for testing. We selected this strategy instead of k-fold cross-validation due to the considerable computational cost associated with training each gradient boosting model.

To ensure unbiased evaluation, the held-out test set was isolated from all training and tuning stages. From the remaining 80% of the data, we created an internal split (80% training / 20% validation) with stratification to preserve the original class distribution. The validation set was used exclusively for hyperparameter selection, thus avoiding leakage in the final evaluation.

The hyperparameters of XGBoost, CatBoost, and LightGBM were tuned with a unified metaheuristic search (Aquila Optimizer). Instead of setting model-specific heuristics, we standardized the hyperparameter search space as follows:

- Depth (`max_depth`): 2–12
- Boosting iterations (`n_estimators`): 50–5000
- Learning rate (`learning_rate`): 0.05–0.50

Within this space, Aquila explored candidate configurations and selected those that maximized the chosen validation metrics (accuracy and macro-F1) under a fixed evaluation cost function. After identifying the best configuration per model, we retrained the corresponding final model on the entire training set (the full 80%) and reported performance only on the untouched test set.

As a summary of the methodology, Fig. 3 presents the workflow of the proposed system for triage classification from clinical text. We initiated the process by preprocessing the dataset, which involved removing stopwords, converting the text to lowercase, lemmatizing, and tokenizing the 'motivo_de_consulta' field. Next, we removed uncategorized records and vectorized the text using a Word2Vec model that had been previously trained and optimized with the Aquila Optimizer algorithm. To address class imbalance, we applied SMOTE oversampling in combination with custom class weight assignment. We then split the resulting dataset using a hold-out validation strategy, allocating 80% for training and 20% for testing. We further divided the training subset to perform hyperparameter tuning. Finally, we trained three gradient boosting models—XGBoost, CatBoost, and LightGBM—and evaluated their performance on the independent

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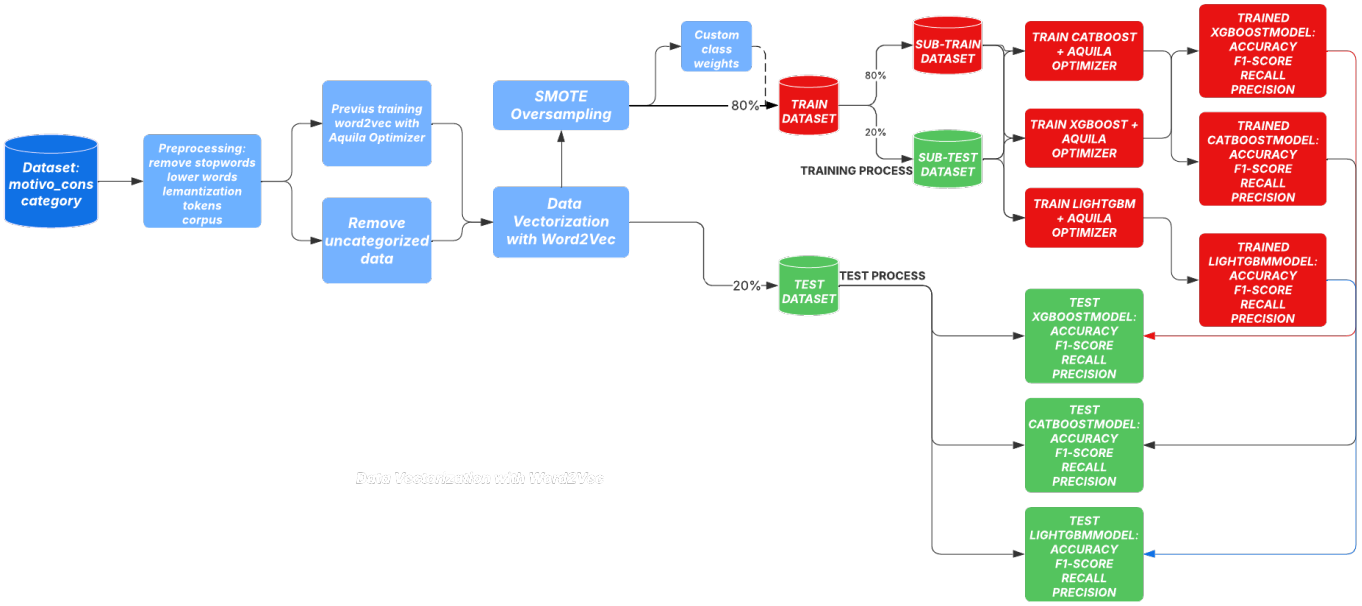


Fig. 3. Workflow for triage classification using Word2Vec embeddings, SMOTE oversampling, and class weighting. Models (XGBoost, CatBoost, LightGBM) are optimized with Aquila and evaluated on a hold-out test set.

test set using Accuracy, F1-score, Recall, and Precision as evaluation metrics.

D. Metrics

We evaluated the performance of the models using multiple metrics suited for multiclass classification, with particular attention to the challenges posed by class imbalance. The selected metrics—Accuracy, Recall, and F1-score—offer complementary perspectives on classification effectiveness and robustness.

- Accuracy (Acc) refers to the overall proportion of correctly classified instances across all classes. While it provides a general measure of performance, it can be misleading in imbalanced datasets, as the majority of classes may dominate the results.
- Recall (macro and weighted) evaluates the model's ability to identify all instances of each class correctly. The macro-average recall (recall(m)) calculates the recall independently for each class and then averages them, treating all classes equally. In contrast, the weighted-average recall (recall(w)) computes the average recall, taking into account the support (number of instances) for each class, thus giving more importance to frequent classes.
- F1-score (macro and weighted) provides a harmonic mean between precision and recall, which is particularly informative in imbalanced scenarios. The macro-average F1-score (F1-Score(m)) gives equal weight to each class, while the weighted F1-score (F1-Score(w)) adjusts the contribution of each class according to its prevalence.

V. Results and Discussion

This section presents the main results obtained after training and evaluating the three gradient boosting algorithms analyzed: XGBoost, CatBoost, and LightGBM. We trained each model using the training set and assessed its performance exclusively with test data.

Table I summarizes the performance metrics used to measure the effectiveness of each model, including accuracy, recall, and both macro-averaged (m) and weighted (w) F1-score of the three gradient boosting models evaluated in this study: XGBoost, LightGBM, and CatBoost. Macro-averaged and weighted F1-score values, as well as recall, are presented to provide a comprehensive view of the behavior of each model in a multiclass problem with unbalanced classes.

The Catboost model achieved the best overall performance, obtaining the highest values in both the weighted F1 score (0.7315) and weighted recall score (0.7334), as well as macro F1 (0.5093). These results indicate that CatBoost successfully maintained balanced classification despite class imbalance. XGBoost ranked second, with macro and weighted F1 score values of 0.7216 and 0.5080, respectively. Although it ranked below CatBoost, it maintained competitive and stable performance on both metrics, in addition to exhibiting greater stability in minority classes. LightGBM, on the other hand, showed similar performance in macro recall (0.4971) and also obtained slightly lower F1 scores, especially in macro (0.5003), suggesting lower efficiency in classification accuracy in dominant classes. Overall, these results confirm that CatBoost offered the best balance between performance in minority and majority classes, consolidating itself as the most robust alternative for addressing the multiclass triage classification problem in this study.

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TABLE I
COMPARISON OF BOOSTING MODELS: WEIGHTED AND MACRO METRICS

Model	F1-score(w)	F1-score(m)	Recall(m)	Recall(w)
CatBoost	0.7315	0.5093	0.5210	0.7334
XGBoost	0.7216	0.5080	0.5000	0.7301
LightGBM	0.7155	0.5003	0.4971	0.7242

Table II shows the performance of the XGBoost model in predicting each triage class. The model demonstrated the best ability to correctly identify critical cases (C1) with an F1 score of 0.4686, indicating that, despite the class imbalance, it accurately identifies a significant proportion of high-severity patients. The class that showed the best performance was C3 (F1 score of 0.7668), followed by class C2, suggesting that the model adequately handles most classes and intermediate classes. However, class C5, despite the model having the highest score in this class, still has a low F1 score (0.1310), which is noticeable in the following models. This is due to the significant imbalance in this class. This behavior is consistent with the usual difficulty of discriminating between intermediate levels of urgency, given their greater clinical ambiguity. Overall, XGBoost provided a solid balance between precision and sensitivity at most triage levels.

TABLE II
PER-CLASS PERFORMANCE METRICS FOR THE XGBOOST MODEL. RESULTS ARE REPORTED FOR EACH TRIAGE CATEGORY USING REAL-WORLD EMERGENCY DEPARTMENT DATA.

Class	Precision	Recall	F1-Score
C1	0.4505	0.4881	0.4686
C2	0.7390	0.8434	0.7878
C3	0.7668	0.7357	0.7509
C4	0.4837	0.3434	0.4017
C5	0.2459	0.0893	0.1310

Table III presents the results by class for the CatBoost model. This model achieved the highest average F1 scores for the majority classes, C2 (0.7938) and C3 (0.7628), with acceptable recall in C1 (0.5238) and C4 (0.4184). However, accuracy decreased significantly in class C5, with a score of 0.1789, indicating a lower ability to identify these cases correctly. In addition, performance in C1 was also lower compared to XGBoost, with an F1 score of 0.4378 and reduced accuracy (0.3761). These results indicate that CatBoost is the ideal model for the most frequently requested cases in the clinical spectrum emergency system; however, its performance is less stable than that of other models in minority classes.

TABLE III
PER-CLASS PERFORMANCE METRICS FOR THE CATBOOST MODEL. THE EVALUATION IS BASED ON INDIVIDUAL CLASS OUTCOMES USING ACTUAL TRIAGE DATA.

Class	Precision	Recall	F1-Score
C1	0.3761	0.5238	0.4378
C2	0.7911	0.7965	0.7938
C3	0.7607	0.7649	0.7628
C4	0.4275	0.4184	0.4229
C5	0.1789	0.1012	0.1293

Table IV reports the results by class for the LightGBM model. This model outperformed the others in class C1, with the highest F1 score among the three models (0.4719), and also achieved the highest recall in C2 (0.8571). However, like the other approaches, LightGBM underperformed in class C5, obtaining the lowest F1 (0.1057) and recall (0.0714) among the models. In addition, a slight decrease in F1 was observed in C3 and C4, with scores of 0.7418 and 0.3963, respectively. Although it is the worst-performing model of the three evaluated, it could be considered a viable option for implementation in the emergency system, given that it achieves acceptable rates in C2 and C3, the most relevant classes.

TABLE IV
PER-CLASS PERFORMANCE METRICS FOR THE LIGHTGBM MODEL. METRICS REFLECT THE PREDICTIVE QUALITY FOR EACH TRIAGE CLASS.

Class	Precision	Recall	F1-Score
C1	0.4468	0.5000	0.4719
C2	0.7256	0.8571	0.7859
C3	0.7734	0.7127	0.7418
C4	0.4667	0.3443	0.3963
C5	0.2034	0.0714	0.1057

In general, the results show a consistent pattern: the three models classify the majority triage classes (C2 and C3) with greater accuracy, which is clinically relevant because these classes account for the majority of patients in the emergency system. Although C1 is relevant, it is atypical for a patient with C1 characteristics to enter the system; generally, these patients are admitted as soon as possible. The most significant difficulty for all models is in class C5, where scores drop sharply due to the strong imbalance and greater clinical ambiguity. Although this class is not treated in the emergency system, the fact that the models have difficulty recognizing it allows for bias in an operational scenario, generating false negatives that could unnecessarily overload the emergency system.

Among the models, CatBoost stood out for its best overall performance (higher weighted F1, weighted recall, and highest macro F1), maintaining a balance between majority and minority classes. XGBoost ranked second, with similar metrics and stable performance, especially in the intermediate classes (C2–C3). LightGBM, meanwhile, obtained the best F1 in C1 and the highest recall in C2, although it performed poorly in C3–C4 and was least effective in C5.

These findings indicate that CatBoost offers the best balance for the evaluated multiclass triage problem, while XGBoost is a solid alternative due to its stability in the intermediate classes. LightGBM remains viable when sensitivity is prioritized in C1–C2, but its performance is weaker in C5.

VI. CONCLUSION

This study demonstrated the feasibility of using natural language processing (NLP) techniques and reinforcement learning algorithms to perform effective triage categorization in an emergency department based on the reason for consultation. The Word2Vec model, optimized using the Aquila optimization algorithm, enabled the semantic representation

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of clinical texts, facilitating their integration into classifiers such as XGBoost, CatBoost, and LightGBM. The results showed that CatBoost achieved the best overall performance, outperforming the other models in both weighted F1 score and macro and recall, positioning it as a solid option for clinical applications. Although minority classes have a low or poor score, they are not the initial priority of the emergency system, since in the most extreme cases (C1), they are treated immediately, with triage rarely being performed, and in less urgent cases (C5), which are generally treated at family health centers (CESFAM). Furthermore, this suggests that Gradient Boosting approaches, combined with embeddings, may be a viable tool to support the prioritization of medical care. However, their clinical utility depends on additional validation processes and careful integration into operational workflows.

The model currently has significant limitations, including the fact that the data are derived from a single high-complexity public hospital in Chile. Therefore, as future work, we recommend using more corpora related to the emergency system in Chile to achieve greater accuracy and improve results.

Future work should complement the preprocessing of the text by incorporating medical dictionaries that allow for the expansion of acronyms and clinical language models, enabling richer semantics, better understanding, embedding, and categorization of the proposed model.

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